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Multimodal Media Retrieval and Captioning System

Group – 15

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1. Abstract

This project presents the design and implementation of a complete vision-language system for two closely related multimodal tasks: multimodal retrieval and automatic image captioning. Both tasks are built end-to-end using PyTorch, trained and evaluated on the Flickr8k and Flickr30k benchmark datasets, and demonstrated through an interactive web interface.

The retrieval component learns a joint embedding space in which semantically related images and natural-language descriptions are projected close together while unrelated pairs are pushed apart. A text query can retrieve the most relevant images from a large collection, and conversely, a query image can retrieve its most relevant captions. The system begins with a pretrained CLIP baseline that requires no training and then progressively builds self-trained contrastive models using ResNet and ViT image encoders paired with LSTM, attention-augmented LSTM, and Transformer text encoders. Retrieval performance is measured with Recall@1, Recall@5, Recall@10, and Mean Reciprocal Rank on both datasets.

The captioning component addresses the generative problem of producing a natural-language description of an image from scratch, word by word. Four architectures of increasing sophistication are developed: a ResNet encoder with an LSTM decoder, the same framework augmented with visual attention, a Vision Transformer encoder with an LSTM decoder, and a ViT encoder paired with a GPT-style autoregressive decoder. Caption quality is evaluated using BLEU, METEOR, ROUGE-L, CIDEr, and SPICE. The progressive model roadmap mirrors real-world research progression, providing both engineering intuition and measurable performance milestones at each stage.

2. Introduction

Imagine uploading a photograph of a golden retriever playing in a snowy park and asking a computer: 'Find me all images similar to this one,' or 'Describe what is happening in this photograph.' A decade ago, such requests required either exact keyword matches or a human being. Today, machines can attempt both tasks — but only because of transformative advances in multimodal artificial intelligence that jointly model images and natural language.

We live in a world saturated with visual content. Platforms such as Google Photos, Instagram, Adobe Stock, and hospital imaging archives collectively host hundreds of billions of images. Locating a specific image using a text description — or automatically generating a textual label for an unlabelled image — has enormous practical value across accessibility software for visually impaired users, medical image reporting, e-commerce product search, autonomous vehicle perception logs, and creative tools for journalists and designers.

2.1 Why Is This Needed?

Traditional image retrieval depends on user-supplied tags or file names. This breaks down when images are untagged, when the query vocabulary differs from stored tags, or when the user wants to search by scene meaning rather than keyword. For example, a query such as 'a child enjoying ice cream at the beach' returns nothing if the only tag on the relevant image is 'summer

outing.' Similarly, users with visual impairments cannot benefit from purely visual content without accurate, automatically generated descriptions.

2.2 Current Gaps

Despite the success of large pretrained models such as CLIP (Radford et al., 2021), several practical and pedagogical gaps remain:

- Large pretrained models are opaque black boxes — practitioners use them without understanding how contrastive training, shared embedding spaces, or autoregressive decoding work internally.
- Most tutorials rely on off-the-shelf APIs, leaving students without the engineering intuition needed to build, debug, or extend these systems.
- Retrieval and captioning are treated as entirely separate problems, yet they share the same image encoder and benefit from joint evaluation frameworks.
- Evaluation beyond top-1 accuracy — Recall@5, CIDEr, SPICE — is rarely covered in a single educational end-to-end system.

2.3 How This Project Helps

This project bridges the gap between using pretrained models and deeply understanding them. By implementing every component from first principles — image preprocessing, vocabulary construction, tokenisation, contrastive loss, LSTM decoders, attention mechanisms, ViT patch embeddings, GPT-style generation, and all evaluation metrics — the work yields both a functional system and transferable engineering knowledge. The eight-phase roadmap progresses from a zero-shot CLIP baseline to advanced ViT+GPT architectures, with clear benchmarks at each stage.

3. Project Objectives

The primary goal is to build a complete, reproducible multimodal vision-language system using PyTorch, implementing every component from the ground up. The specific objectives are:

1. Implement end-to-end data pipelines in PyTorch: dataset loading, image preprocessing, text tokenisation, vocabulary creation, padding, and dataloader construction for Flickr8k and Flickr30k.
2. Build multimodal retrieval models from scratch using contrastive learning — from a pretrained CLIP zero-shot baseline through self-trained ResNet+LSTM, ResNet+LSTM+Attention, ViT+LSTM, and ViT+Transformer models trained with bidirectional InfoNCE contrastive loss.
3. Build image captioning models from scratch: ResNet+LSTM, ResNet+LSTM+Attention, ViT+LSTM, and ViT+GPT decoder — learning teacher forcing, cross-entropy loss, and autoregressive decoding.
4. Acquire all core technical concepts from first principles: embedding spaces, cosine similarity, contrastive loss, attention mechanisms, sequence generation with <SOS>/<EOS> tokens, padding masks, and beam search.

5. Evaluate models rigorously: retrieval models with Recall@1/5/10 and Mean Reciprocal Rank; captioning models with BLEU-1/4, METEOR, ROUGE-L, CIDEr, and SPICE — reported independently for Flickr8k and Flickr30k so the effect of dataset scale is measurable.
6. Deploy an interactive Flask or Streamlit demo where users can upload images, enter text queries, select models, and view retrieval results or generated captions in real time.

4. Literature Review

The field of vision-language learning has evolved rapidly from early statistical co-occurrence models to large-scale transformer-based foundation models pretrained on billions of image-text pairs. The works surveyed below span more than a decade of progress and directly inform the architecture, training methodology, dataset choices, and evaluation protocols of this project.

4.1 Show and Tell: A Neural Image Caption Generator (Vinyals et al., 2015)

Vinyals et al. introduced one of the first fully end-to-end neural image captioning systems. A deep CNN (GoogLeNet) encodes the image into a fixed-length feature vector, which initialises the hidden state of an LSTM that generates the caption word by word. The entire model is trained jointly by maximising the log-likelihood of correct next-word predictions — a regime now known as teacher forcing. At the time of publication, Show and Tell achieved state-of-the-art BLEU scores on Flickr30k and MS-COCO, demonstrating that end-to-end gradient descent could replace hand-crafted image description pipelines. This paper directly motivates the first captioning baseline (ResNet+LSTM) implemented in this project.

4.2 Deep Visual-Semantic Alignments (Karpathy & Fei-Fei, 2015)

Karpathy and Fei-Fei proposed a model that aligns fragments of image regions with fragments of natural-language sentences using a multimodal recurrent neural network. Rather than representing an image as a single global vector, they extracted CNN features from bounding-box regions and learned region-word compatibility scores. This is a conceptual precursor to cross-modal attention. Crucially for this project, they evaluated on Flickr8k, Flickr30k, and MS-COCO using Recall@1, Recall@5, and Recall@10, establishing the retrieval evaluation protocol replicated here.

4.3 Show, Attend and Tell (Xu et al., 2015)

Xu et al. extended Show and Tell by introducing a soft spatial attention mechanism. Instead of compressing the entire image into a single vector, the CNN produces a spatial feature map. At each decoding step the LSTM computes attention weights for every spatial location, forming a context vector as a weighted sum of spatial features. When generating 'dog', attention peaks over the dog region; when generating 'sky', it shifts to the background. The model achieved significant BLEU and METEOR improvements and established attention as an essential captioning component. This work directly motivates the Stage C2 captioning model (ResNet+LSTM+Attention).

4.4 Microsoft COCO Captions Benchmark (Chen et al., 2015)

Chen et al. released the MS-COCO captioning dataset and an automated evaluation server that standardised caption quality measurement. The benchmark defined the five metrics now universally used: BLEU, METEOR, ROUGE-L, CIDEr, and SPICE. It also established the protocol of collecting five human reference captions per image — the same protocol followed by Flickr8k and Flickr30k. Although this project trains on Flickr datasets, the COCO evaluation toolkit is used to compute all captioning metrics, ensuring comparability with published results.

4.5 CLIP: Contrastive Language-Image Pre-Training (Radford et al., 2021)

Radford et al. trained dual-encoder models on 400 million (image, alt-text) pairs using a contrastive objective that maximises the similarity of matching pairs and minimises that of non-matching pairs within each training batch. CLIP achieved zero-shot ImageNet classification accuracy comparable to a fully supervised ResNet-50 and set new state-of-the-art results on MS-COCO and Flickr30k retrieval benchmarks. The CLIP contrastive loss (InfoNCE / bidirectional cross-entropy over the batch similarity matrix) and the dual-encoder retrieval paradigm are the direct foundation for all five retrieval models built in this project.

4.6 An Image is Worth 16×16 Words — ViT (Dosovitskiy et al., 2020)

Dosovitskiy et al. demonstrated that a pure transformer architecture applied to non-overlapping 16×16 image patches matches or surpasses CNN performance on image classification when pretrained on large datasets. Each patch is linearly projected into an embedding, a CLS token is prepended, and the sequence is processed by standard multi-head self-attention blocks. Unlike CNNs, ViT captures long-range spatial dependencies from the first layer. ViT is used as the image encoder in the advanced retrieval (R4–R5) and captioning (C3–C4) models in this project.

4.7 GPT-2: Language Models are Unsupervised Multitask Learners (Radford et al., 2019)

Radford et al. introduced GPT-2, a large autoregressive language model based on the transformer decoder with masked self-attention. When adapted for image captioning, visual features from ViT are injected as a conditioning prefix or via cross-attention, and the GPT decoder generates the caption token by token. This architecture — ViT + GPT-style decoder — forms the most advanced captioning model in this project (Stage C4) and approximates modern systems such as BLIP and Flamingo.

4.8 Summary of Related Works

Paper	Year	Key Contribution	Relevance to Project
Show and Tell (Vinyals et al.)	2015	End-to-end CNN+LSTM captioning; teacher forcing	Stage C1 captioning baseline
Deep Visual-Semantic Alignments (Karpathy & Fei-Fei)	2015	Region-word alignment; Recall@K evaluation	Retrieval evaluation protocol

Show, Attend and Tell (Xu et al.)	2015	Soft spatial attention over CNN feature maps	Stage C2 attention captioning
MS-COCO Captions (Chen et al.)	2015	Benchmark: BLEU / CIDEr / SPICE metrics	Evaluation metric definitions
CLIP (Radford et al.)	2021	Contrastive dual-encoder; zero-shot retrieval	Retrieval baseline + InfoNCE loss
ViT (Dosovitskiy et al.)	2020	Patch-based transformer image encoder	Stages R4–R5; C3–C4
GPT-2 (Radford et al.)	2019	Autoregressive transformer decoder	Stage C4 captioning model

5. Dataset Description

5.1 Flickr8k

Flickr8k (Hodosh et al., 2013) contains 8,000 images from Flickr, each annotated with exactly five independent human-written English captions, yielding 40,000 image-caption pairs. Images depict everyday scenes with people, animals, and objects in varied indoor and outdoor settings. The dataset is small enough for rapid prototyping and debugging on a single GPU, making it the primary development dataset for all model stages.

Split	Images	Captions	Purpose
Training	6,000	30,000	Model parameter optimisation
Validation	1,000	5,000	Hyperparameter tuning and early stopping
Test	1,000	5,000	Final evaluation (held out until evaluation phase)
Total	8,000	40,000	—

5.2 Flickr30k

Flickr30k (Young et al., 2014) extends the Flickr concept to approximately 31,783 images, each with five captions, yielding ~158,915 pairs. The larger scale tests whether model performance improves with more training data. The identical pipeline developed on Flickr8k is applied to Flickr30k by changing a single configuration parameter.

Split	Images	Captions	Purpose
Training	~29,000	~145,000	Model parameter optimisation
Validation	~1,014	~5,070	Hyperparameter tuning
Test	~1,000	~5,000	Final evaluation
Total	~31,783	~158,915	—

5.3 COCO (Microsoft Common Objects in Context)

MS-COCO (Lin et al., 2014) is the standard benchmark for image captioning, with over 330,000 images and five captions each (~1.5 million captions). While this project trains on Flickr datasets, the COCO evaluation toolkit and its metric definitions (BLEU, METEOR, ROUGE-L, CIDEr, SPICE) are adopted to ensure comparability with the broader published literature.

5.4 Top 5 Representative Samples — Flickr8k

	Image	Caption 1	Caption 2	Caption 3	Caption 4	Caption 5
0	Sample_0	A black dog is running after a white dog in th...	Black dog chasing brown dog through snow	Two dogs chase each other across the snowy gro...	Two dogs play together in the snow .	Two dogs running through a low lying body of w...
1	Sample_1	A little baby plays croquet .	A little girl plays croquet next to a truck .	The child is playing croquette by the truck .	The kid is in front of a car with a put and a ...	The little boy is playing with a croquet hamme...
2	Sample_2	A brown dog in the snow has something hot pink...	A brown dog in the snow holding a pink hat .	A brown dog is holding a pink shirt in the snow .	A dog is carrying something pink in its mouth ...	A dog with something pink in its mouth is look...
3	Sample_3	A brown dog is running along a beach .	A brown dog wearing a black collar running acr...	A dog walks on the sand near the water .	Brown dog running on the beach .	The large brown dog is running on the beach by...
4	Sample_4	A black and white dog with a red Frisbee stand...	A dog drops a red disc on a beach .	A dog with a red Frisbee flying in the air .	Dog catching a red Frisbee .	The black dog is dropping a red disc on a beach .

5.5 Top 5 Representative Samples — Flickr30k

	Image	Caption 1	Caption 2	Caption 3	Caption 4	Caption 5
0	1000092795.jpg	Two young guys with shaggy hair look at their ...	Two young , White males are outside near many ...	Two men in green shirts are standing in a yard .	A man in a blue shirt standing in a garden .	Two friends enjoy time spent together .
1	10002456.jpg	Several men in hard hats are operating a giant...	Workers look down from up above on a piece of ...	Two men working on a machine wearing hard hats .	Four men on top of a tall structure .	Three men on a large rig .
2	1000268201.jpg	A child in a pink dress is climbing up a set o...	A little girl in a pink dress going into a woo...	A little girl climbing the stairs to her playh...	A little girl climbing into a wooden playhouse	A girl going into a wooden building .
3	1000344755.jpg	Someone in a blue shirt and hat is standing on...	A man in a blue shirt is standing on a ladder ...	A man on a ladder cleans the window of a tall ...	man in blue shirt and jeans on ladder cleaning...	a man on a ladder cleans a window
4	1000366164.jpg	Two men , one in a gray shirt , one in a black...	Two guy cooking and joking around with the cam...	Two men in a kitchen cooking food on a stove .	Two men are at the stove preparing food .	Two men are cooking a meal .

5.6 Dataset Directory Structure

Folder / File	Contents
data/flickr8k/images/	All 8,000 JPEG image files
data/flickr8k/captions.csv	image_id, caption_id, caption_text, split columns
data/flickr8k/splits/	train.txt, val.txt, test.txt — image-level split lists
data/flickr30k/images/	All ~31,783 JPEG image files
data/flickr30k/captions.csv	Same schema as Flickr8k
data/flickr30k/splits/	train.txt, val.txt, test.txt

6. Model Architecture

The system comprises two parallel tracks — Retrieval and Captioning — that share the same image preprocessing pipeline and diverge in how encoded visual features are used downstream.

6.1 High-Level System Overview

Block	Purpose	Main Outputs
Dataset preparation	Download, organise, clean, and split Flickr8k and Flickr30k.	Train/validation/test image-caption pairs.
Image preprocessing	Convert raw images into normalised tensors for all encoders.	Image tensors of shape $B \times 3 \times H \times W$.
Text preprocessing	Clean captions, tokenise, build vocabulary, pad sequences.	Token IDs, attention masks, vocabulary files.
Model training	Train retrieval and captioning models stage by stage.	Model checkpoints and evaluation metrics.
Demo interface	Allow users to select dataset, task, and model; run inference.	Interactive Flask or Streamlit application.

6.2 Retrieval Models — Five Progressive Stages

All retrieval models project image and text inputs into a shared d -dimensional embedding space ($d = 256$ or 512). Embeddings are L2-normalised and compared with cosine similarity. The InfoNCE contrastive loss is computed over an $N \times N$ batch similarity matrix where diagonal entries are correct (positive) pairs and all off-diagonal entries serve as in-batch negatives.

Stage	Image Encoder	Text Encoder	Training Objective
R1 – Pretrained CLIP	CLIP ViT-B/32 (frozen)	CLIP Transformer (frozen)	Zero-shot cosine similarity; no training.
R2 – ResNet + LSTM	ResNet-50 (ImageNet)	Embedding + 1-layer LSTM	Bidirectional InfoNCE contrastive loss.
R3 – ResNet + LSTM + Attention	ResNet-50	LSTM + Bahdanau attention	Bidirectional InfoNCE contrastive loss.
R4 – ViT + LSTM	ViT-B/16 (pretrained)	Embedding + 1-layer LSTM	Bidirectional InfoNCE contrastive loss.
R5 – ViT + Transformer	ViT-B/16	GPT / Transformer encoder	Bidirectional InfoNCE contrastive loss.

6.3 Captioning Models — Four Progressive Stages

Stage	Image Encoder	Text Decoder	Key Design Feature
C1 – ResNet + LSTM	ResNet-50	Single-layer LSTM	Image feature initialises LSTM hidden state.
C2 – ResNet + LSTM + Attention	ResNet-50	LSTM + Bahdanau visual attention	Decoder attends to spatial feature map at each step.
C3 – ViT + LSTM	ViT-B/16	LSTM decoder	Patch-level ViT features replace CNN spatial features.
C4 – ViT + GPT Decoder	ViT-B/16	GPT-style autoregressive decoder	Cross-attention on visual tokens; closest to modern VLMs.

6.4 Contrastive Loss for Retrieval

For a training batch of N image-caption pairs, an $N \times N$ cosine similarity matrix is computed. Ground-truth assignments are on the diagonal. Two cross-entropy losses are computed: one per image querying over all captions, one per caption querying over all images. The final loss is their average. This bidirectional formulation maximises alignment in both retrieval directions simultaneously.

6.5 Caption Generation at Inference

Method	Description	Trade-off
Greedy decoding	Select highest-probability word at each step.	Fast; may miss the globally best caption.
Beam search ($k=3-5$)	Maintain k candidate sequences at each step.	Slower; typically produces better captions.
Top-k / Top-p sampling	Sample from the top-k or cumulative-p words.	Diverse outputs; reduced factual accuracy.

7. Pre-Processing

7.1 Image Pre-Processing Pipeline

Raw image files are converted into fixed-size, normalised floating-point tensors before being passed to any encoder. The exact transformations depend on whether the encoder is ResNet, ViT, or CLIP, but the overall pipeline is consistent across all stages.

7. Load: read from disk using PIL or OpenCV; convert to RGB (handles grayscale and RGBA).
8. Resize: resize the shorter edge to 256 px while maintaining the aspect ratio.
9. Crop: random 224×224 crop during training; centre 224×224 crop at validation/test.
10. Augmentation (training only): random horizontal flip and colour jitter. No augmentation at eval time.
11. Normalise: scale pixels from $[0, 255]$ to $[0, 1]$, then standardise channel-wise using ImageNet mean $\mu = [0.485, 0.456, 0.406]$ and std $\sigma = [0.229, 0.224, 0.225]$. CLIP-specific values for the CLIP baseline.
12. Batch: stack B images into a tensor of shape $B \times 3 \times 224 \times 224$ (default $B = 32$).

Step	Training	Validation / Test
Resize	Shorter edge $\rightarrow$ 256 px	Shorter edge $\rightarrow$ 256 px
Crop	Random crop to 224×224	Centre crop to 224×224
Augmentation	Random horizontal flip; colour jitter	None
Normalisation	ImageNet (or CLIP) mean & std	Same as training
Output shape	$B \times 3 \times 224 \times 224$	$B \times 3 \times 224 \times 224$

7.2 Text Pre-Processing Pipeline

Text preprocessing converts raw caption strings into padded integer token sequences suitable for embedding layers and attention-based models.

7.2.1 Text Cleaning

13. Convert all text to lowercase ('Dog' and 'dog' → same token).
14. Remove punctuation (retaining apostrophes in contractions) and extra whitespace.
15. Optionally remove rare symbols and normalise unicode characters to ASCII.

7.2.2 Tokenisation

Captions are split into word tokens using whitespace splitting or the NLTK word tokeniser. For transformer models, subword tokenisation (BPE or WordPiece) can be applied. The CLIP baseline uses CLIP's native tokeniser directly.

Example: 'A dog is running in the grass.' → ['a', 'dog', 'is', 'running', 'in', 'the', 'grass']

7.2.3 Special Tokens

Token	ID	Meaning	Use Case
<PAD>	0	Padding token	Pads shorter captions to uniform batch length.
<SOS>	1	Start of sequence	Prepended to every caption; signals start of generation.
<EOS>	2	End of sequence	Appended to every caption; signals end of generation.
<UNK>	3	Unknown word token	Replaces words absent from the training vocabulary.

7.2.4 Vocabulary Construction

16. Collect and tokenise all captions from the training split only — val/test splits are never used.
17. Count word frequencies across the entire training corpus.
18. Retain words appearing at least twice (threshold configurable); discard rare words.
19. Sort retained words alphabetically for reproducibility across runs.
20. Assign fixed IDs to special tokens: <PAD>=0, <SOS>=1, <EOS>=2, <UNK>=3.
21. Assign consecutive IDs starting from 4 to all remaining vocabulary words.
22. Persist word_to_id.json and id_to_word.json for use at training and inference time.

Approximate vocabulary sizes: Flickr8k → 2,500–3,500 tokens; Flickr30k → 7,000–10,000 tokens (both depend on the minimum-frequency threshold).

7.2.5 Numericalization and Padding

Each caption is converted to integer IDs with <SOS> prepended and <EOS> appended. All captions in a batch are right-padded to the length of the longest caption using <PAD>=0.

Caption : 'a dog is running in grass'
Tokens : [<SOS>, 'a', 'dog', 'is', 'running', 'in', 'grass', <EOS>]
IDs : [1, 12, 45, 8, 230, 19, 88, 2]
Padded (10): [1, 12, 45, 8, 230, 19, 88, 2, 0, 0]
Attn mask : [1, 1, 1, 1, 1, 1, 1, 1, 0, 0]

7.2.6 Attention Masks

An attention mask tensor of shape $B \times L$ accompanies every tokenised batch. A value of 1 indicates a real token; 0 indicates padding. Attention masks prevent transformer-based encoders and GPT-style decoders from attending to padding positions, which would corrupt learned representations.

7.3 Dataloader Output Format

Task	Fields Returned by Dataloader
Retrieval	image_tensor ($B \times 3 \times 224 \times 224$), caption_token_ids ($B \times L$), attention_mask ($B \times L$), image_id, caption_id
Captioning	image_tensor, input_caption_ids (teacher-forced input), target_caption_ids (one-step shifted), attention_mask, image_id
CLIP Baseline	CLIP-preprocessed image tensor, CLIP-tokenised text (CLIP's own tokeniser)

8. Evaluation Metrics

8.1 Retrieval Evaluation Metrics

Metric	Definition
Recall@1	Correct match appears at rank 1 (top-ranked result).
Recall@5	Correct match appears within the top 5 ranked results.
Recall@10	Correct match appears within the top 10 ranked results.
Median Rank	Median position of the first correct result; lower is better.
Mean Reciprocal Rank	Average of $1/\text{rank}$ of the first correct result; higher is better.

Both text-to-image and image-to-text retrieval directions are evaluated and reported separately on the Flickr8k and Flickr30k test sets.

8.2 Captioning Evaluation Metrics

Metric	What It Measures
BLEU-1 / BLEU-4	Unigram and 4-gram precision overlap with five reference captions.
METEOR	Synonym-aware recall and precision with morphological stemming.
ROUGE-L	F-measure based on the longest common subsequence.
CIDEr	TF-IDF-weighted n-gram consensus with human captions.
SPICE	Scene-graph semantic matching: objects, attributes, and relations.

Qualitative evaluation is also performed: a random sample of generated captions is manually inspected for object correctness, relational accuracy, fluency, and hallucination.

9. Implementation Timeline

Phase	Deliverable
Phase 1	Dataset download and cleaning; image preprocessing pipeline; text preprocessing, vocabulary, and dataloader.
Phase 2	Pretrained CLIP retrieval baseline on Flickr8k and Flickr30k.
Phase 3	Self-trained ResNet+LSTM (R2) and ResNet+LSTM+Attention (R3) retrieval models.
Phase 4	ViT+LSTM (R4) and ViT+Transformer (R5) retrieval models.
Phase 5	ResNet+LSTM (C1) and ResNet+LSTM+Attention (C2) captioning models.
Phase 6	ViT+LSTM (C3) and ViT+GPT (C4) captioning models.
Phase 7	Comprehensive evaluation of all models; comparative results tables.
Phase 8	Flask or Streamlit demo UI with integrated checkpoints and interactive inference.

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